Kuber Sarwal

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Education

Georgia Institute of Technology — Atlanta, GA

B.S. in Computer Science, Artificial Intelligence and Systems & Architecture — *May 2028*

- **GPA: 3.86**
- **Coursework:** Data Structures and Algorithms, Intro to Artificial Intelligence, Computer Organization and Programming, Discrete Math, Linear Algebra, Differential Equations, Object-Oriented Programming, Calculus III

Experience

Machine Learning Engineering Intern – GNC — Irvine, CA

Terran Orbital Corporation, a Lockheed Martin Company — *Jun. 2026 – Aug. 2026*

- Custom-built a **U-Net CNN** on night-sky imagery to detect star centroids under high-speed maneuvers via **star centroid maps**, achieving **23 arcsecond pointing error** at a **5 deg/s slew rate** – a **3x improvement**.
- Implemented **QUEST** on CNN centroids for real-time **quaternion attitude estimation**; deployed via **TensorRT quantization** and **CUDA/cuDNN** on **NVIDIA Jetson** with **OpenCV** vision, cutting latency **40%**.
- Developed a **synthetic star-streak simulation environment** to test and validate the CNN/QUEST pipeline pre-deployment; validated attitude estimates against IMU telemetry across **12+ hardware-in-the-loop tests.**

Machine Learning Researcher – SAIL Lab — Atlanta, GA

Georgia Institute of Technology — *Aug. 2026 – Present*

- Extended **AHEAD** teleoperation with a **cross-attention fusion** module conditioning intent prediction on language commands, improving accuracy **12%** and enabling joint action classification for multi-robot control.

Machine Learning Researcher – Dynamics and Control Group — Atlanta, GA

Georgia Institute of Technology — *Oct. 2026 – Present*

- Developing a **Res-U-Net** mapping **dual-wavelength interferograms** (460/640 nm) to **thickness profiles**, using the **synthetic wavelength** to resolve **phase wrapping**; reached **18 nm RMSE** on held-out test profiles.
- Built a **physics-based simulator** rendering interferograms from **Gaussian random field** thickness profiles with noise, illumination nonuniformity, and LED bandwidth for **domain randomization**, generating **50K** pairs.

Machine Learning Engineer, VIP – AI MakerSpace Nexus VI — Atlanta, GA

Georgia Institute of Technology (Sponsored by NVIDIA) — *Jan. 2026 – Present*

- Built StudyBuddy with **OCR, NLP, and RAG** on Llama 3/Ollama; orchestrated **LLM workflows** via **LangChain** to generate adaptive modules from quiz scores, with **SQLite** progress tracking and **content tagging**.
- Evaluated **ChatGT** and **CUDA Llama 3** fine-tuning on **NVIDIA PACE ICE cluster**; benchmarked retrieval accuracy/hallucination rate across **50+ datasets**, driving a **70%** improvement in student comprehension.

Programming Projects

AI DJ – Learned Transition Scorer | *Python, PyTorch, LightGBM, librosa* | GitHub — Oct. 2026 – Present

- Built a **learning-to-rank** model scoring downbeat pairs across tracks via a **listwise softmax loss**, learning **Spotify's hand-tuned costs** from **8,287 DJ transitions**; hits DJ cues within **4 bars** on **68%** of test pairs.
- Fine-tuned **Beat This!** on **Serato** beat grids as **pseudo-labels** across **11K+ tracks**, raising bar-accurate downbeats on held-out hand-annotated tracks from **22%** to **88%** (Serato: **93%**) for fully automatic inference.

Depot Trajectory Prediction | *Python, PyTorch, PyG, C++, SQL, ONNX* — Sep. 2025 – Present

- Built a **PyG graph neural network** predicting trajectories of **30+ vehicles** in dense depots, cutting **minADE 42%** vs. a constant-velocity baseline; mined **10K+ logs** in **SQL** to target long-tail failures (**miss rate 31%** to **18%**); deployed in **C++** via **ONNX** at **12 ms**/frame, validated on **2K+** replay scenarios for real-time use.

Technical Skills

Languages: Java, Python, JavaScript, HTML/CSS, SQL, NoSQL, C++, C#, R, Go, Rust, TypeScript, MATLAB
Frameworks: React, Node, Flask, Angular, Django, FastAPI, AWS, Apache Spark, Kafka, LangChain, Spring
Developer Tools: Git, Microsoft Roo Code, Claude code, Azure, MongoDB, CUDA, MLflow, Docker, Kubernetes, MCP
Libraries: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, OpenCV, PyG, ONNX, TensorRT, LightGBM, librosa